SCOPE · VIT CHENNAI

MediTwin

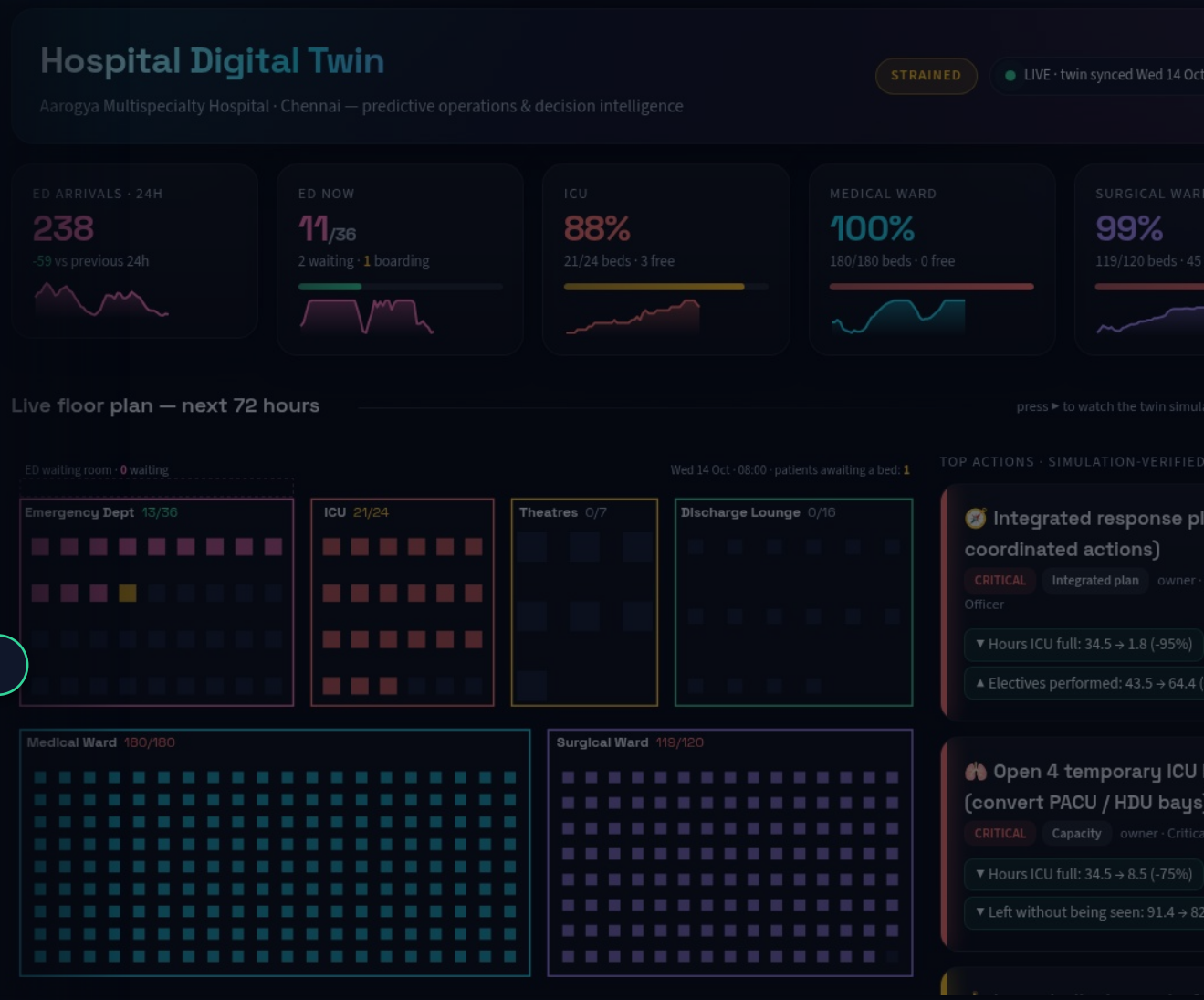
AI-Powered Hospital Digital Twin for Predictive Healthcare Operations and Decision Intelligence

Forecast | Simulate | Optimise | Recommend

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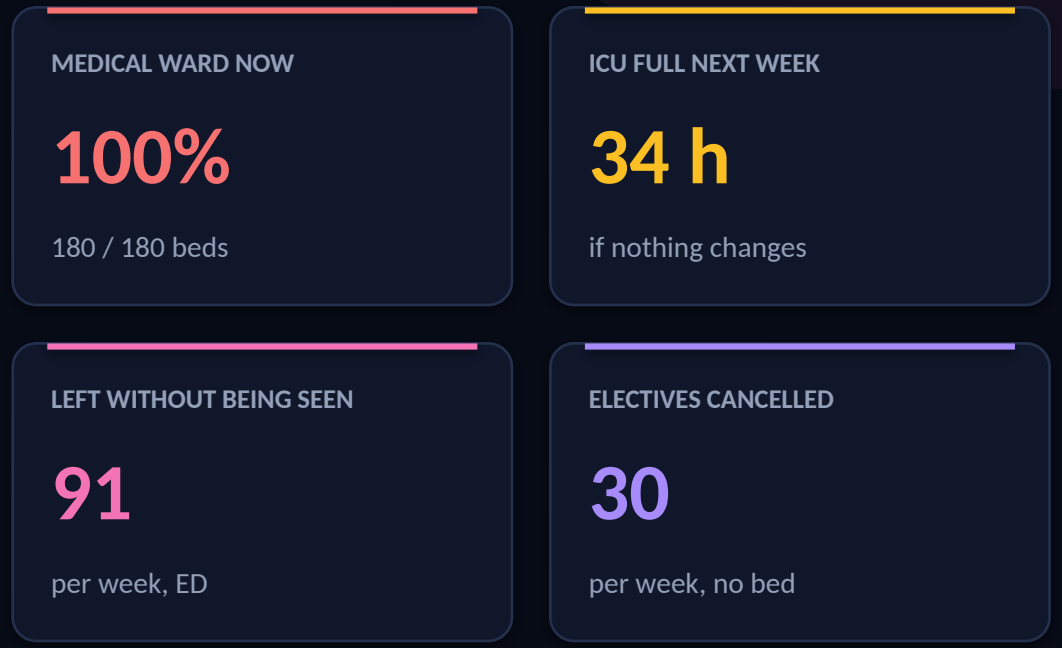
School of Computer Science and Engineering



THE PROBLEM

Hospitals run on yesterday's numbers

- **Reactive operations.** Bed crises, ED crowding and cancelled surgery are noticed only after they happen.
- **Siloed decisions.** Nursing rosters, theatre lists and bed management are planned separately, though they compete for the same beds.
- **No safe place to test.** A new policy (surge beds, diversion, deferring electives) can only be tried on real patients.
- **Fragmented research.** Most work does forecasting, simulation or optimisation, rarely all three, and almost never verifies its advice.



Baseline 7-day projection of the twin from its live state (Wed 14 Oct 2026, 08:00).

OBJECTIVES

What MediTwin sets out to do

01

Forecast

Predict ED arrivals and ICU / ward / ED occupancy 14 days ahead, with calibrated uncertainty and explanations.

02

Mirror

Hold a live digital twin of every bed, queue and patient, and simulate the hospital forward from that exact state.

03

Optimise

Turn forecasts into nurse rosters, theatre lists and patient transfers that respect real constraints.

04

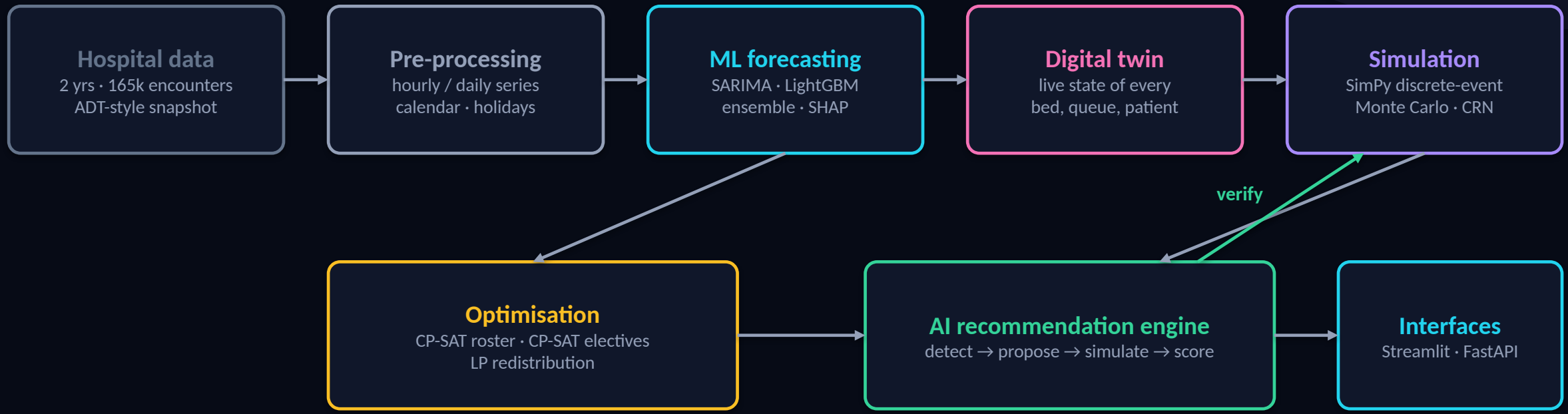
Recommend

Propose actions and test each one in the twin first. Only advice that helps is shown.

Novelty: all four layers in one loop. Every recommendation is re-simulated and reported with its KPI change and 90% confidence interval before anyone sees it.

HOW IT FITS TOGETHER

System architecture



One command builds the full twin in ~72 s: hospital-twin build → generate history → back-test 4 models × 5 targets → fit + explain → optimise → project 24 futures → verify recommendations.

DATA

A realistic synthetic hospital

BEDS	ENCOUNTERS	THEATRES	ESI LEVELS
360	164,594	7	5
ED 36 · ICU 24 · Med 180 · Surg 120	730 days of history	+ 16-chair discharge lounge	triage, LWBS patience, boarding

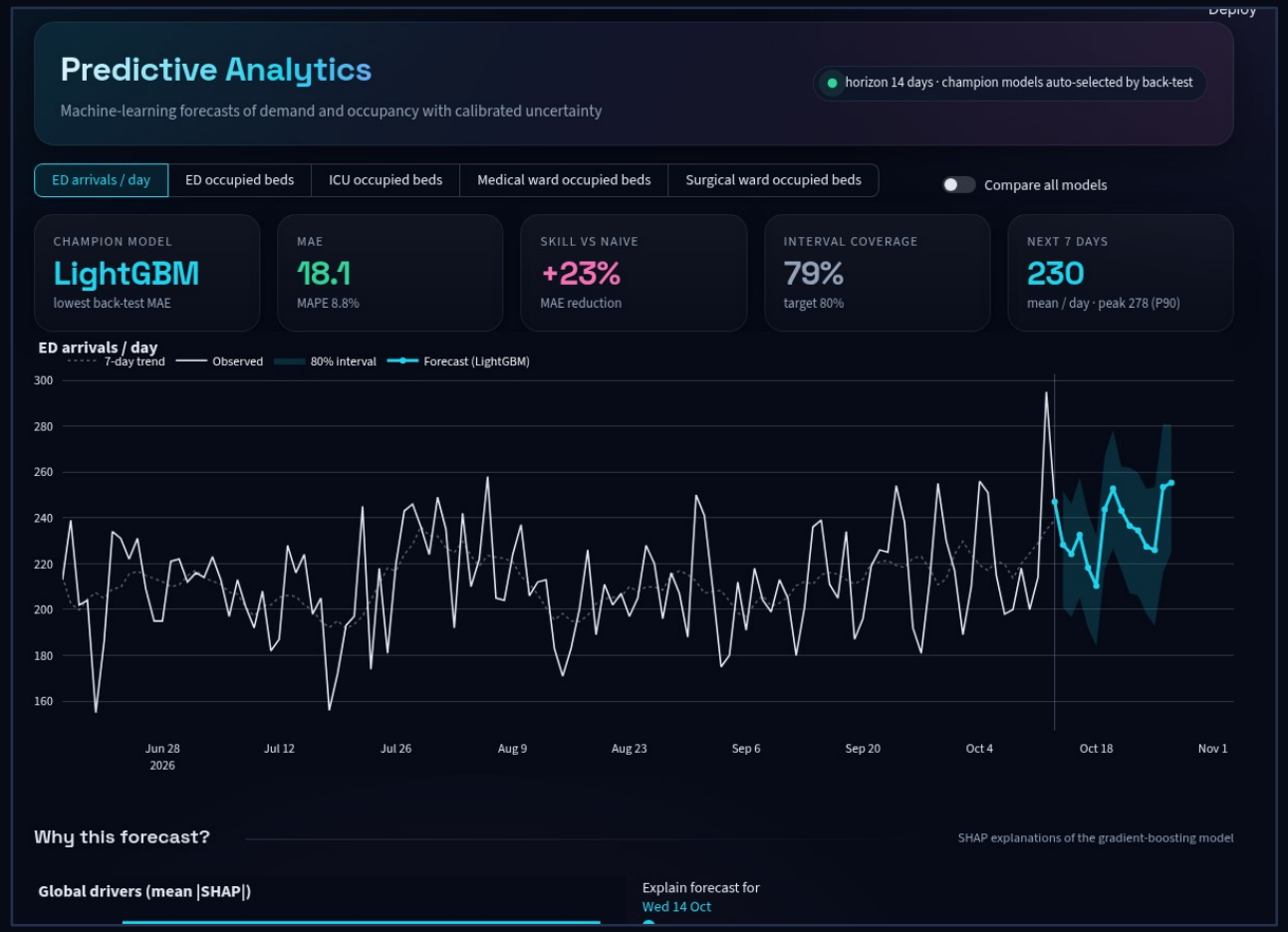
Demand reflects Chennai

- Weekly cycle and hour-of-day profile
- North-east monsoon fever season, summer heat, winter respiratory wave
- Indian public holidays and a Diwali trauma spike
- AR(1) day-to-day noise and random mass-casualty events

Why it is credible

- History is produced by running demand through the simulator, so occupancy, boarding and cancellations respect real capacity limits
- Fictional hospital (Aarogya Multispecialty, Chennai): no patient data used
- Real HL7 / FHIR ADT feeds would replace the generator without other changes

Forecasting with calibrated uncertainty



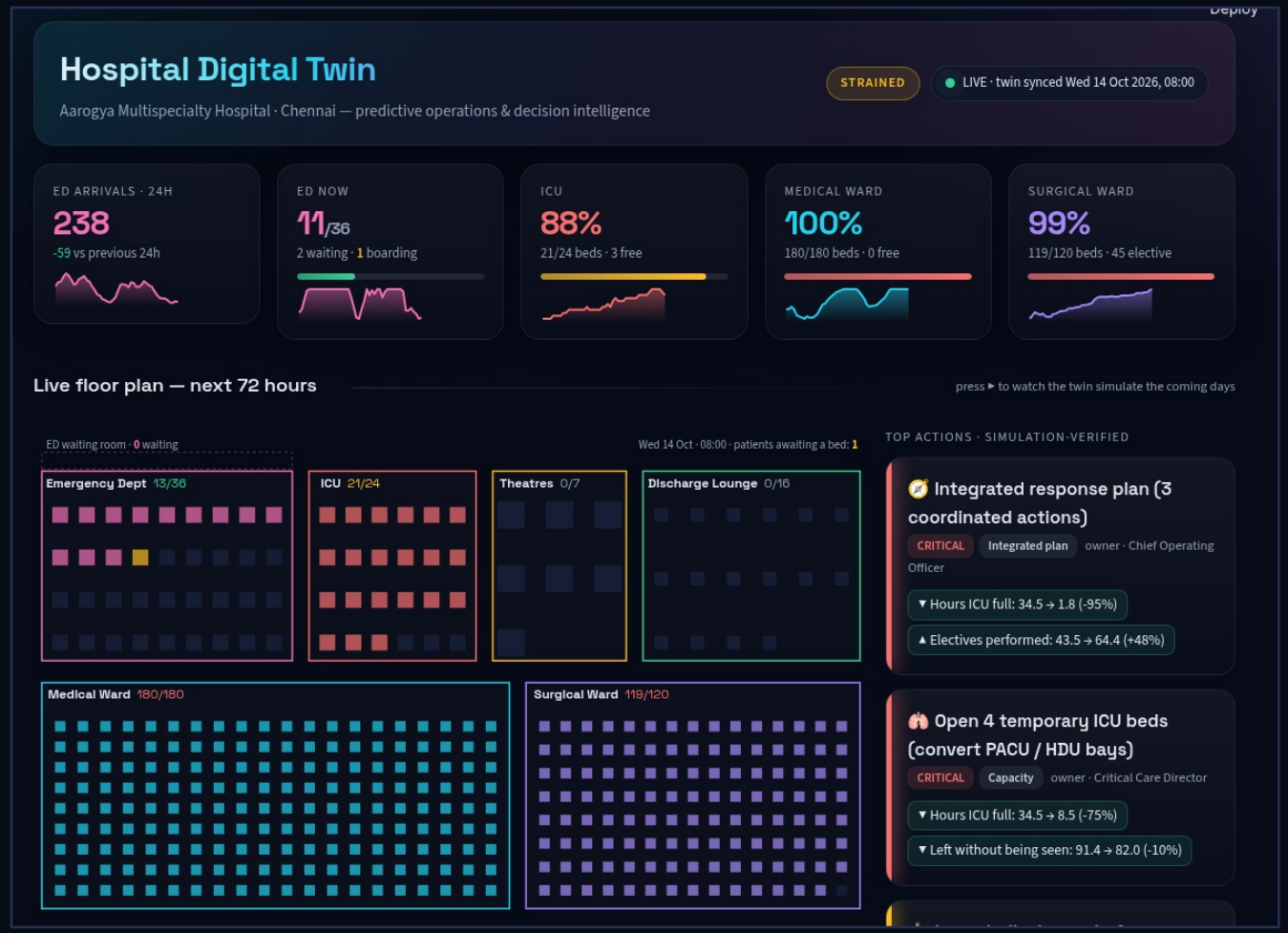
- **4 models:** seasonal naive, SARIMA with holiday regressors, recursive LightGBM, ensemble
- **Rolling back-test:** 10 origins × 14-day horizon picks a champion per target
- **80% bands** calibrated on back-test residuals
- **SHAP** explains the drivers of every forecast day

Target	Champion	MAPE	Skill	Cover
ED arrivals	LightGBM	8.8%	+23%	79%
ED census	LightGBM	10.6%	+28%	81%
ICU census	Ensemble	13.2%	+18%	79%
Medical ward	Ensemble	6.5%	+19%	82%
Surgical ward	LightGBM	9.9%	+24%	81%

Skill = MAE reduction vs seasonal naive. Cover = actuals inside the 80% band.

DIGITAL TWIN + SIMULATION

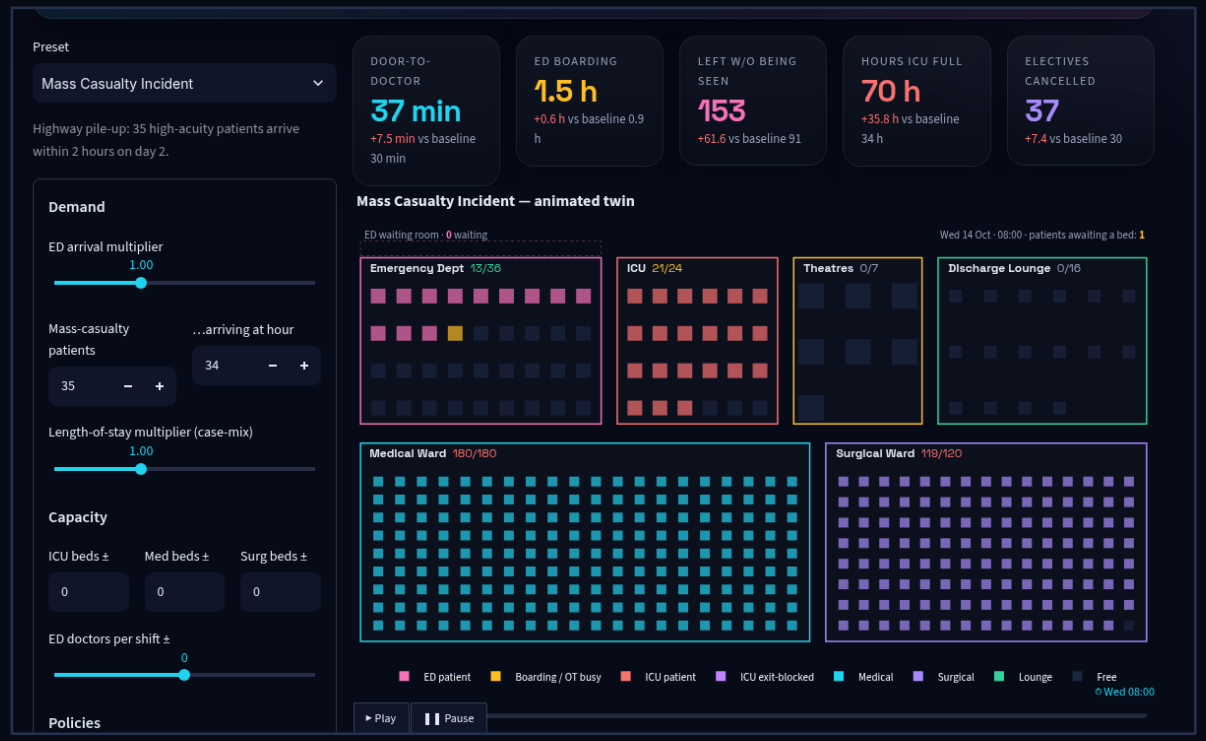
The live digital twin

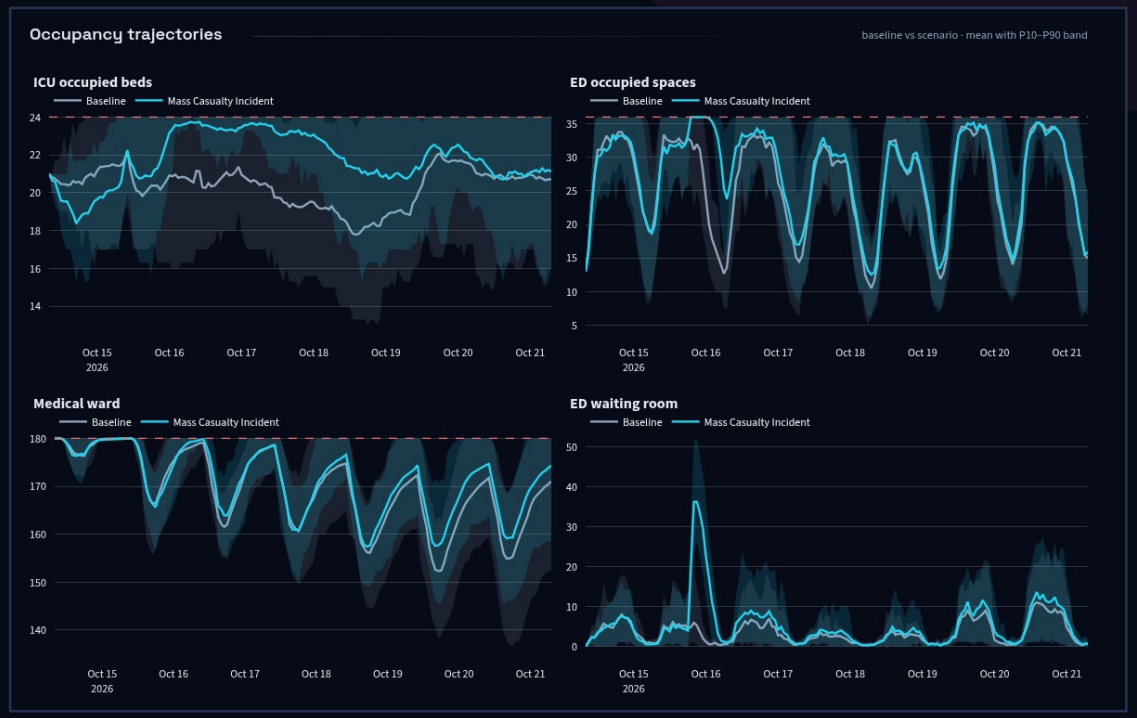


- **Snapshot** of every bed, queue and patient: unit, elapsed stay, acuity
- **Warm start:** simulations begin from now, sampling remaining LOS from truncated log-normals
- **SimPy model:** ESI triage, LWBS, doctor shifts, boarding and outliers, emergency and elective theatres, ICU exit-block, discharge window, lounge
- **Monte Carlo:** 24 futures, demand drawn from the forecast band, common random numbers for fair comparisons
- **Animated floor plan** plays the next 72 h bed by bed

WHAT-IF SIMULATION

Scenario Lab: stress-test before it happens

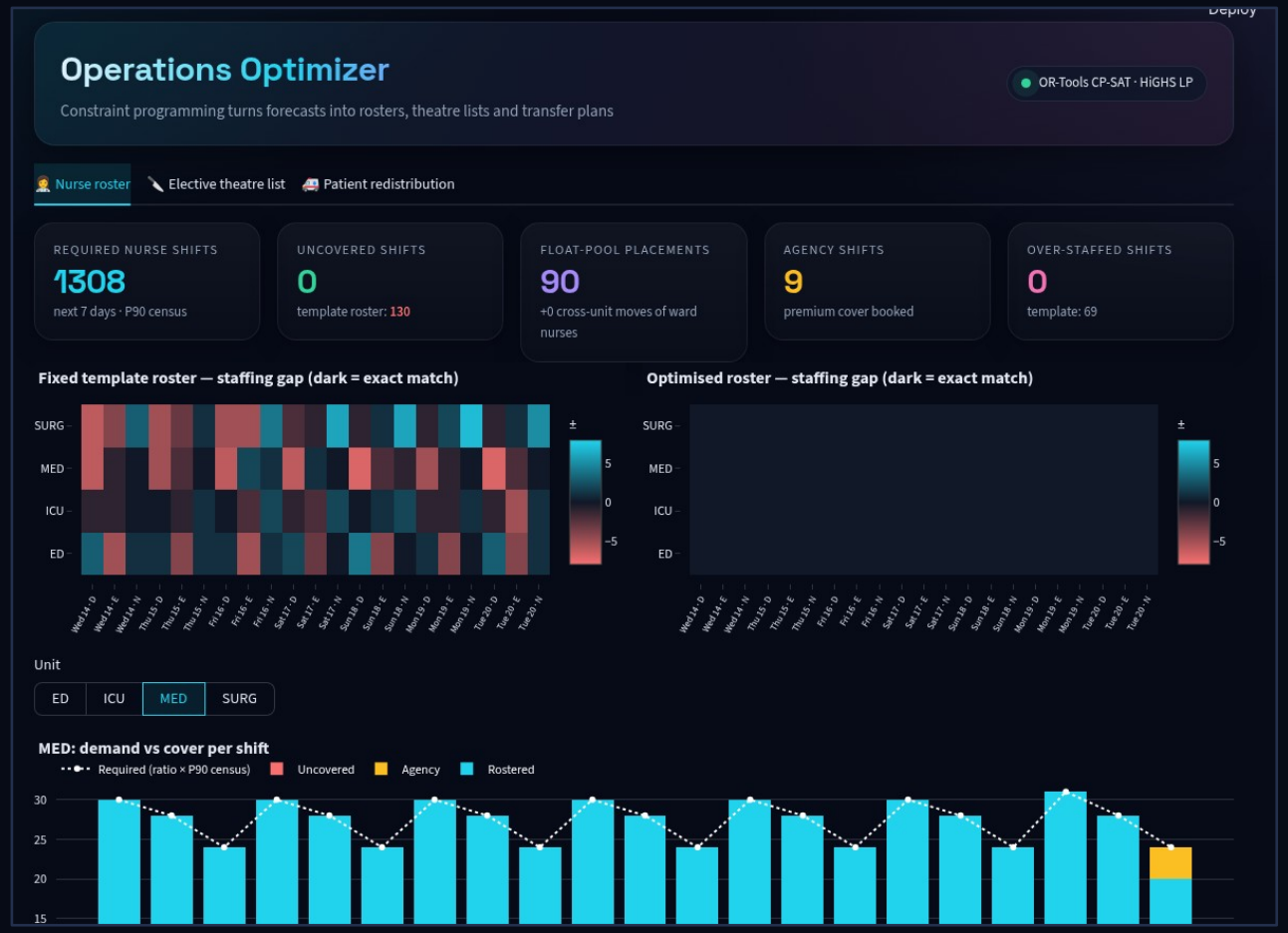




Mass casualty preset: 35 high-acuity patients arrive 2 hours into day 2. Door-to-doctor rises to 37 min and ICU is full for 70 h. Other presets: monsoon fever surge, ICU bed closure, staff shortage, festival week. One click generates an AI response plan for the scenario.

OPERATIONS RESEARCH

Optimisation: from forecast to plan



Nurse roster

OR-Tools CP-SAT

≤ 1 shift / day, no night → day, contracted hours, night cap, float pool, agency

0

uncovered shifts

template: 130

Elective list

OR-Tools CP-SAT

theatre hours + post-op ICU and ward beds against forecast emergency load

0

bed-overflow days

FIFO: 11

Redistribution

SciPy LP (HiGHS)

moves excess patients to outlier beds or partner hospitals at minimum cost

LP

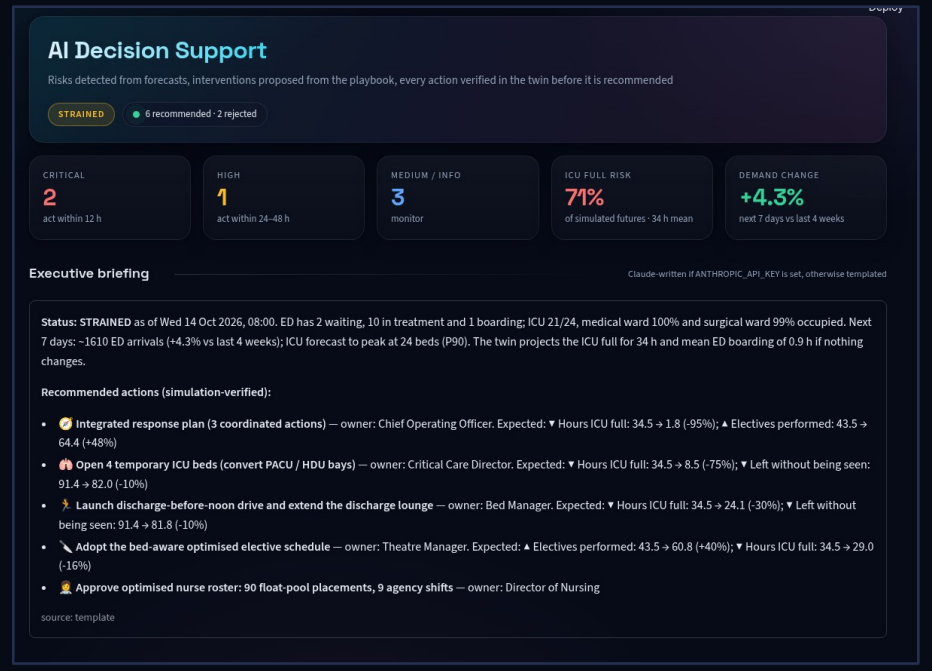
optimal transfers

DECISION INTELLIGENCE

AI recommendations, verified by simulation

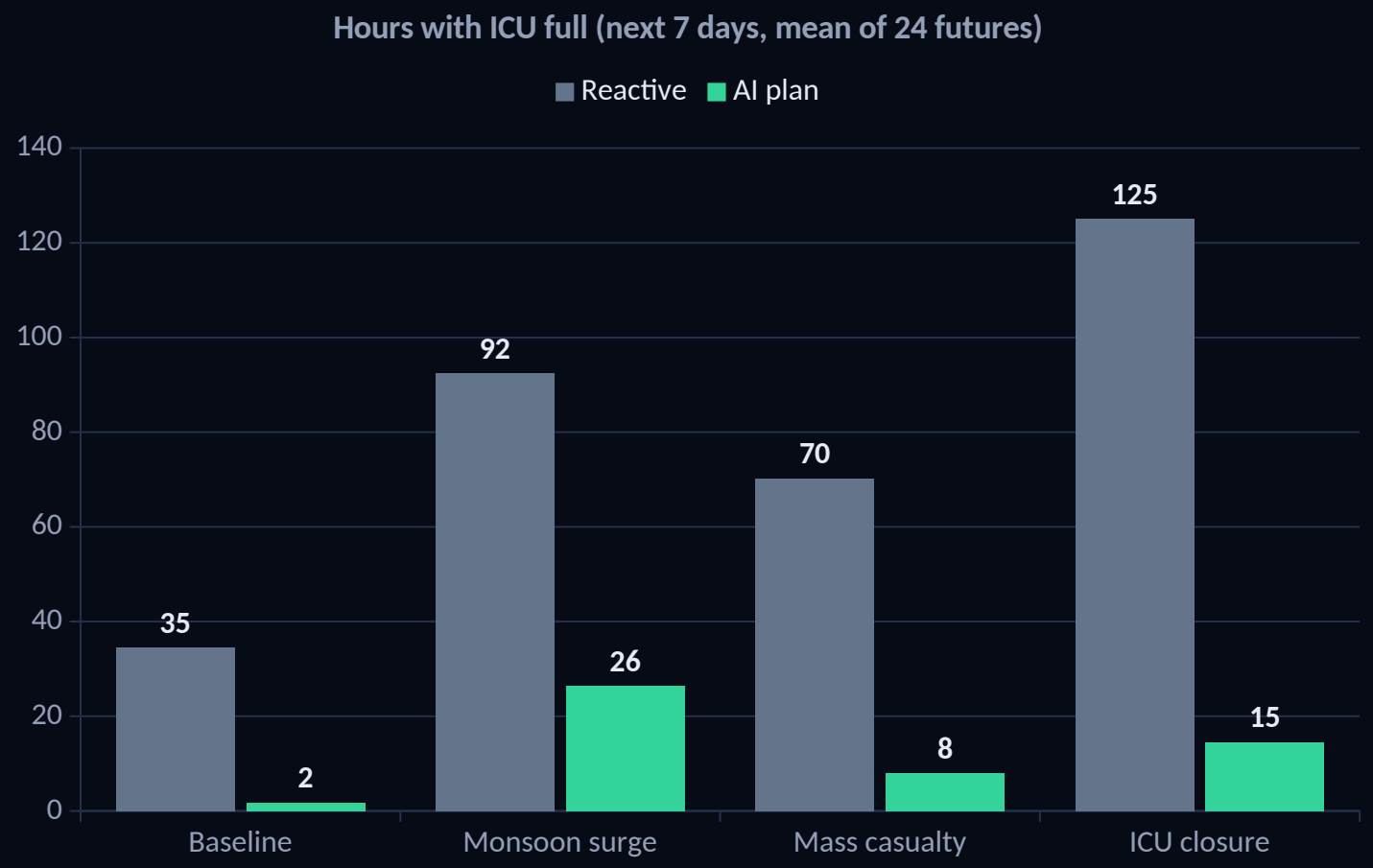
Detect risks in forecasts and twin projections → **Propose** playbook actions with owner and urgency → **Simulate** re-run the twin with each action (CRN) → **Score** weighted KPI benefit with 90% CI → **Bundle** combine winners and re-verify as one plan

Verdict	Action	Simulated effect	Score
Accept	Integrated response plan	ICU full 34.5 → 1.8 h	185
Accept	Open 4 temporary ICU beds	ICU full −75%	85
Accept	Discharge-before-noon drive	ICU full −30%	85
Accept	Bed-aware elective schedule	electives +40%	73
Accept	Optimised nurse roster	0 uncovered shifts	18
Reject	Extra ED physician	cost > benefit	−17
Reject	Defer 30% of electives	lost surgery outweighs gain	−51



ABLATION

Results: AI plan vs reactive operations



Under stress	Reactive	AI plan
Door-to-doctor (MCI)	37 min	**19 min**
ED boarding (ICU closure)	2.3 h	**0.4 h**
LWBS (monsoon)	606	**245**
LWBS (ICU closure)	172	**24**
Electives cancelled (MCI)	37	**21**

3.5–19×

fewer ICU-full hours across every stress scenario, with shorter ED waits and fewer patients leaving unseen.

VALIDATION

Can we trust the twin?

Twin replay: last 8 weeks of observed demand

Series	Observed	Twin	MAPE
ED arrivals / h	8.92	8.95	6.1%
ED census	22.9	23.2	8.2%
ICU census	19.5	20.3	14.1%
Medical ward	153.3	156.0	5.0%
Surgical ward	89.1	94.7	10.2%

Optimisers vs simple rules

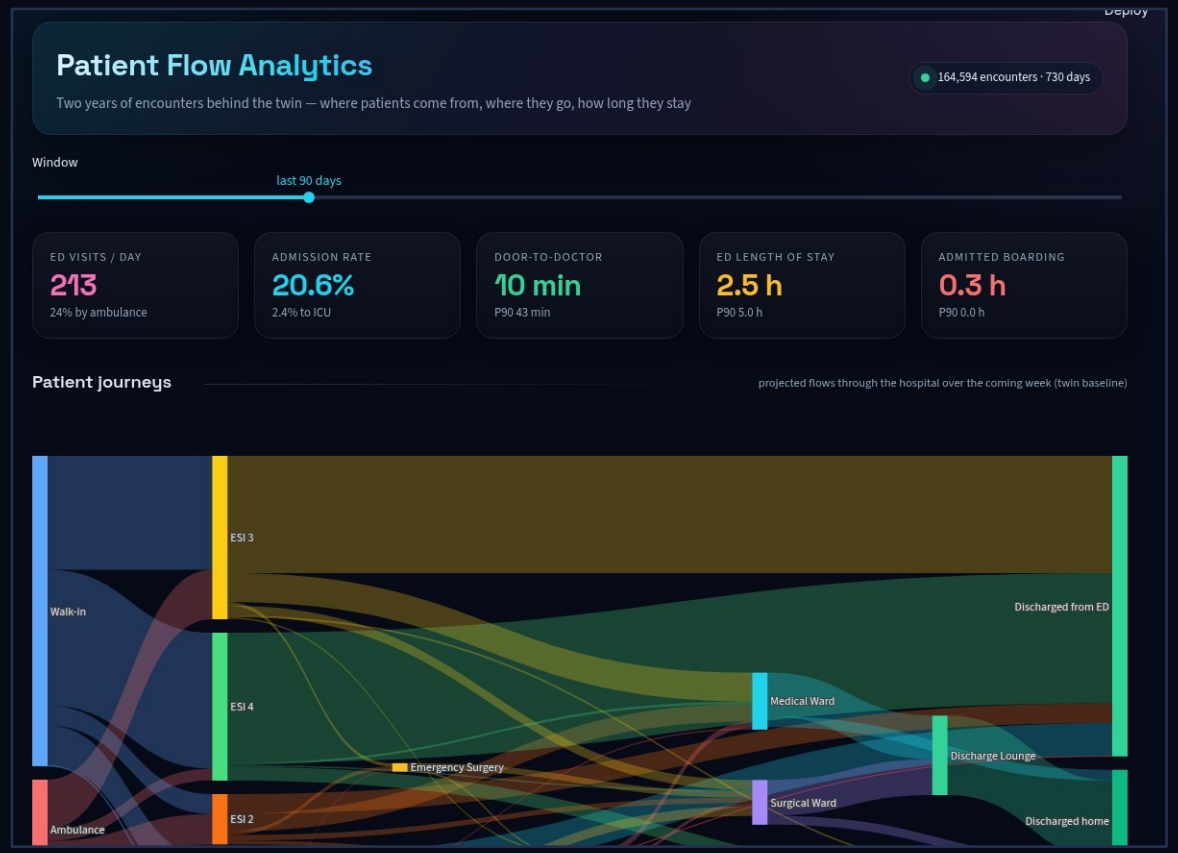
Metric	Rule-based	Optimised
Uncovered nurse shifts (7 d)	130	0
Electives: bed-overflow days	11	0
Electives: P1 cases scheduled	77%	94%

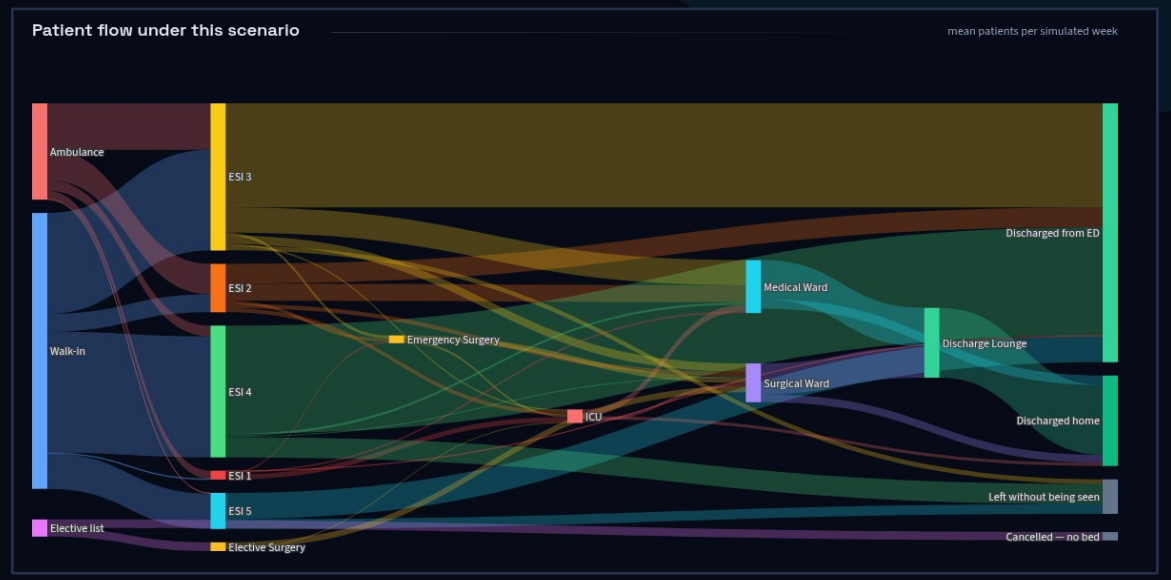


Model Lab page: architecture, back-test detail, twin validation, optimisation benchmarks and ablation, regenerated by hospital-twin evaluate.

INTERFACES

The command centre





REST API (FastAPI): GET /state · /forecast/{target} · /roster · /electives
POST /simulate · /recommendations

Command Center | Forecasts | Scenario Lab | Optimizer | AI Recommendations | Patient Flow | Model Lab

IMPLEMENTATION

Technology stack

Language & build	Python 3.12 · uv · pytest (17 tests)
Forecasting	statsmodels SARIMAX · LightGBM · SHAP · scikit-learn
Simulation	SimPy discrete-event · joblib parallel Monte Carlo
Optimisation	Google OR-Tools CP-SAT · SciPy linprog (HiGHS)
Decision layer	simulation-verified playbook · optional Claude-written executive brief
Interfaces	Streamlit · Plotly (animated floor plan, Sankey, maps) · FastAPI

uv run hospital-twin build · evaluate · app · api

NEXT STEPS

Limitations and future work

Limitations

- Patient data is synthetic. Volumes are plausible for an Indian tertiary hospital, but it is not a real record.
- Twin validation is partly self-consistent, because history comes from the same simulator.
- Nurse staffing is optimised but does not slow care inside the simulator (ED doctors do).
- Partner-hospital capacity is static and fictional.

Future work

- Connect live HL7 / FHIR ADT feeds and validate against a partner hospital.
- Model nurse workload in the simulator, so roster gaps affect patient flow.
- Reinforcement learning for sequential bed-management policies.
- Live regional bed availability for inter-hospital transfers.

Thank you

Forecast → Simulate → Optimise → Recommend

github.com/Arnav717/hospital-digital-twin

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